

PepHAR: Anchor-Based Autoregressive Peptide Generation

Step 0: Initial state — peptide length L=12, receptor given

EBM density model scores receptor surface → identify hotspot locations

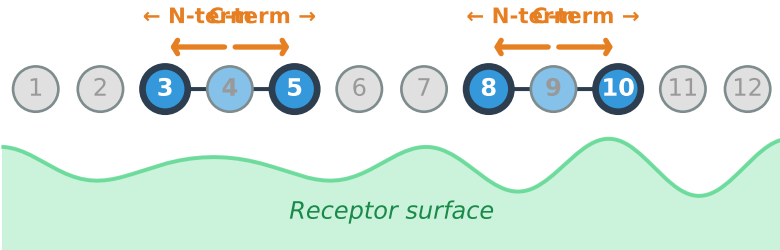


Step 1: Generate K anchor (hotspot) residues via EBM sampling

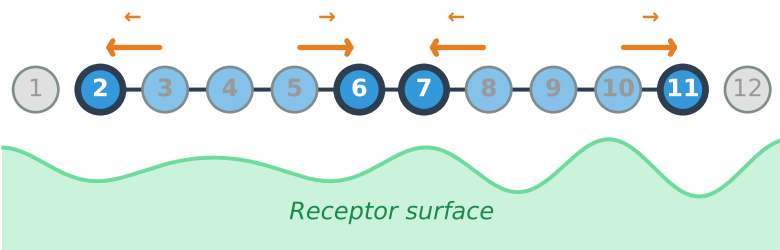
Anchor positions chosen by even spacing | EBM optimizes (coords, AA type)



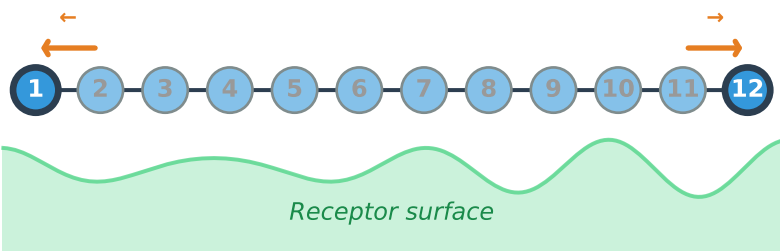
Step 2: Bidirectional extension — first round (1 residue each direction per anchor)



Step 3: Continue bidirectional extension — round 2



Step 4: Fill remaining termini — fragments merge into complete chain



Step 5: Complete peptide — optional joint fine-tuning of all residues

Fine-tune: jointly optimize all coords + dihedral angles + AA identities

